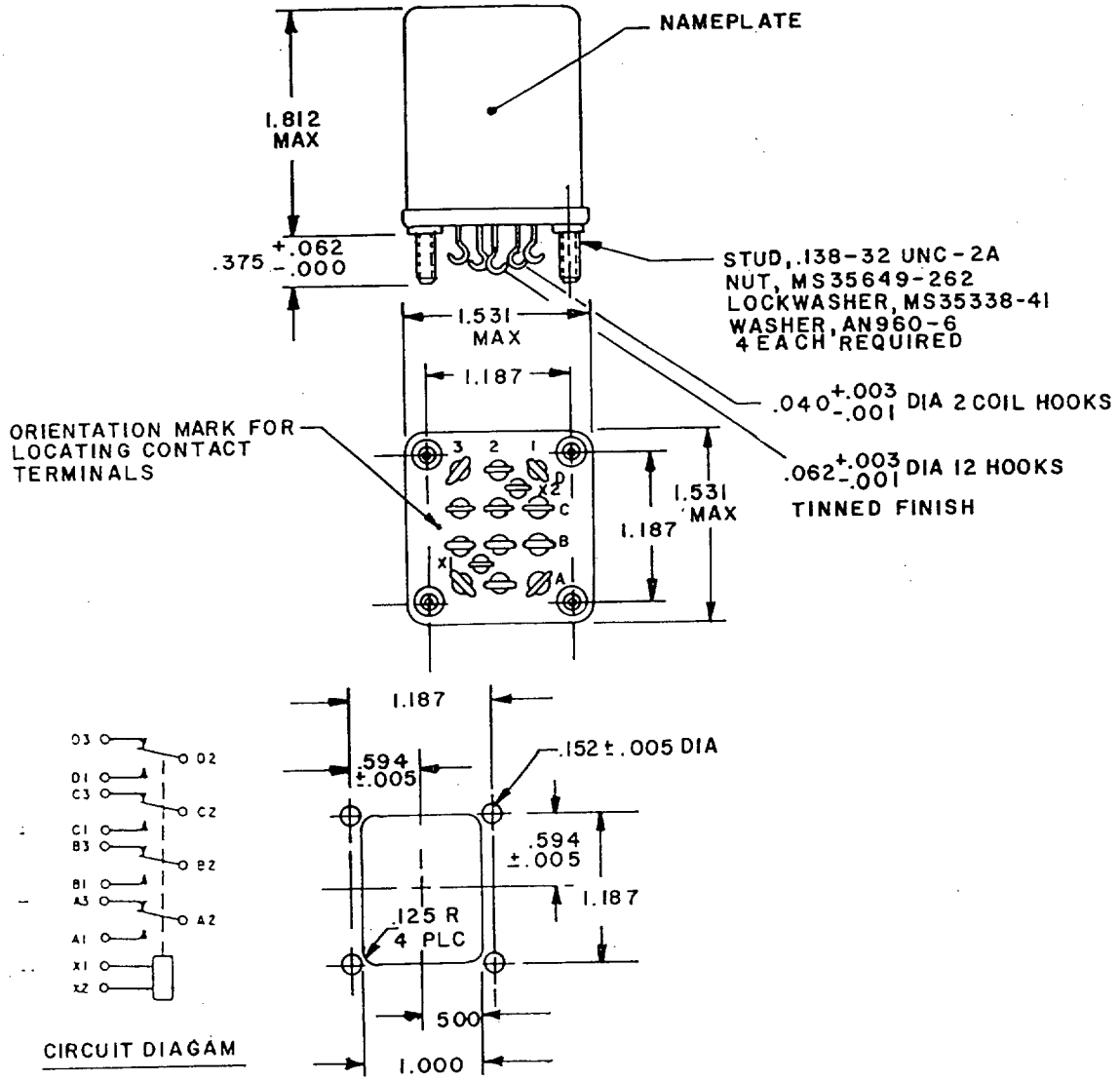
FED. SUP CLASS
5945



G ENTIRE STANDARD REVISED

APPROVED 17 March 1958 REVISED G 5 JUN 87

User activities: Army - Navy - Air Force -

Review activities: Army - EC Navy - Air Force - 11, 99

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P.A	International interest	TITLE	MILITARY STANDARD
USAF - 85 Other Cust Navy - AS		RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4 PDT, TYPE I, SOLDER HOOK HERMETICALLY SEALED	MS25271
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 5

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AMSC N/A

5945-0745-07

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Inches	mm	Inches	mm
.001	0.03	.152	3.86
.003	0.08	.375	9.53
.005	0.13	.500	12.70
.010	0.25	.594	15.09
.040	1.02	1.000	25.40
.062	1.57	1.187	30.15
.125	3.18	1.531	38.89
.138	3.51	1.812	46.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified herein.
5. In the event of confilict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
6. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

Dash number MS25271-	Type	Coil	Terminal type	Mounting	Max weight in pounds
D1	I	dc	Solder hook	Stud	.45
A1	I	ac	Solder hook	Stud	.45

User activities: Army - Navy - Air Force -
Review activities: Army - Navy - EC Air Force - 11, 99
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Review activities: Army -
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Air Force - 11, 99

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Air Force -

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USAF - 85
Other Cust
Navy - AS

International interest

Procurement Specification
MIL-R-6106

SUPERSEDES:

TITLE
RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4 PDT, TYPE I, SOLDER HOOK, HERMETICALLY SEALED

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TABLE II. Operating characteristics.

MS part no. MS25271-	Coil data											Time (milliseconds max)					
	Coil	Nominal			Max		Max pick-up voltage			Drop-out voltage 2/	Hold voltage 2/	Operate 3/	Release 4/	Contact bounce			
														Main		Aux	
		Volts 1/	Freq. Hz	Res Ω ±10%	Volts	Amperes	Normal 2/	High temp test	Cont current test					NO	NC	NO	NC
D1	X1, X2	28	dc	100	29	0.350	18	19.5	22.5	1.5	7.0	20	20	2	2		
A1	X1, X2	115	400 5/	N/A	122	0.073	90	95	103	5.0	30	25	50	2	2		

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ MS25271-A1 may be used on 60 Hz if maximum ambient temperature is +85°C (coil current shall be 0.077 ampere maximum).

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TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10	6			10	6			
Inductive	100													
Inductive	20	6	6			6	4			6	4			
Motor	100	4	4			4	3			4	3			
Lamp	100	2	2			2	1.5			2	1.5			
Transfer load														2/
Mechanical life reduced current	400	2.5	2.5			2.5	2			2.5	2			
Intmd current	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 G
Duration	11 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	10 G
Frequency range	5 - 1,500 Hz
Vibration - random	
Applicable specification	N/A
Power spectral density	N/A
RMS G min	N/A
Frequency range	N/A
Curve	N/A
High shock	N/A
Acceleration	15 G

Electrical characteristics

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms

Dielectric strength (sea level)

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts		
All other points	1,500 V rms	1,125 V rms

Dielectric strength (altitude)

	80,000 ft
Coil to case	250 V rms
Aux contacts	
All other points	350 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	20 amperes
Rupture current	25 amperes
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

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